A Generalized Vacuum-Λ FLRW Solution and the Sign Convention of k

1. Setup

Starting line element:

$ds^2 = A(t,r)\, dt^2 - a(t)^2 B(r)\, dr^2 - a(t)^2 C(r)\, r^2\, d\theta^2 - a(t)^2 C(r)\, r^2 \sin^2(\theta)\, d\varphi^2$

Computing the Ricci tensor (independently verified via direct Christoffel/Riemann summation, no tensor packages) gives four non-zero components R00, R01, R11, R22.

R01 gives:

$R01 \propto (da/dt)\, (\partial A/\partial r) / A = 0$

Since $da/dt \neq 0$ (non-static universe), this forces $\partial A/\partial r = 0$, i.e. **A(t,r) = A(t)**, confirmed independently.

2. Reduced field equations

With A=A(t), the system reduces to three equations (R00, R11, R22) in three unknowns A(t), B(r), C(r), satisfying $R_{\mu\nu} = \Lambda g_{\mu\nu}$ (vacuum + cosmological constant).

3. Solving R00 for A(t)

Solving R00 for A(t) introduces an integration constant k. The correct general solution (confirmed by direct symbolic substitution) is:

$A(t) = 3\, (da/dt)^2 / (\Lambda\, a(t)^2 + 3|k|)$

The absolute value is essential — without it, the formula only reproduces the correct on-shell behavior for one sign of k. The two single-sign candidates,

- $3(da/dt)^2/(\Lambda a^2 - 3k)$ ("standard" form, valid for $k \leq 0$)
- $3(da/dt)^2/(\Lambda a^2 + 3k)$ (valid for $k \geq 0$)

are each individually consistent with R00 reducing to 0=0 — but that check alone is weak: R00 alone does not uniquely fix A(t) for an arbitrary a(t). It only closes once a(t) additionally satisfies its own first-integral (Friedmann) constraint, and that constraint's sign is exactly what differs between the two candidates. Two formulas that are mirror images under $k \to -k$ will always both individually "check out" this way — that is not yet evidence they describe different physics.

4. Solving for B(r), C(r): the general solution

Plugging A(t) (with |k|) into R11 and R22 and solving algebraically:

B(r) = (r C'(r) + 2 C(r))^2 / [4 C(r) (r^2 |k| C(r) + 1)]

with **C(r) left as a completely free function**.

This was verified directly: substituting A(t) (with |k|) and this B(r) (with |k|) into both R^1_1 and R^2_2, with C(r) kept symbolic, both equations reduce to a pure constant identity — independent of r and of the choice of C(r). This is a genuine, exactly verified solution of the field equations for arbitrary C(r), not a special case.

Full solution:

- A(t,r) = 3(da/dt)^2 / (Λa(t)^2 + 3|k|)
- B(t,r) = a(t)^2 (rC'+2C)^2 / [4C(r^2|k|C+1)]
- $\mathcal{N}$(t,r) ≡ angular metric factor = a(t)^2 C(r)
- k ∈ $\mathbb{R}$ free; C(r) free

This reduces to the standard FLRW metric when C(r)=1, k ≤ 0. For k>0, the C(r)=1 reduction gives B(r) = 1/(1+kr^2), differing from the standard RCPC form B(r)=1/(1−kr^2).

5. Resolving the k>0 discrepancy: curvature invariant check

To settle whether B(r)=1/(1+kr^2) (this work) and B(r)=1/(1−kr^2) (standard) for the *same* k represent different physics or merely different conventions, the intrinsic Ricci scalar of the spatial 3-metric (C=1 in both cases — a coordinate-invariant quantity, unaffected by any remaining coordinate freedom) was computed directly:

- Standard metric (B = 1/(1−kr^2)): **R_3 = +6k**
- This work's metric (B = 1/(1+kr^2)): **R_3 = −6k**

Since the Ricci scalar is a curvature invariant, this is a rigorous, coordinate-independent proof: a solution labeled "k" in this work has the *same actual spatial curvature* as the standard solution labeled "−k". The two families are not different physics — they are the same solution space under the relabeling **k → −k**.

6. Conclusion

- The generalized solution with free C(r) is a correctly-derived, independently-verified exact solution of the vacuum+Λ field equations.
- It is **not** evidence of an error in the standard FLRW equations.
- The apparent disagreement at k>0 is fully accounted for by an internal sign convention: this work's k is the negative of the standard curvature parameter.

- The genuinely useful/interesting result is the explicit demonstration — via a fully worked, independently checked exact solution — that the sign of the FLRW curvature parameter k is a convention, not a physically forced choice, together with an explicit free function C(r) generalizing the radial gauge.

Appendix A: Notes on conventions used

- Metric signature: (+,−,−,−), with $R_{\mu\nu} = \Lambda g_{\mu\nu}$ as the vacuum+Λ field equation form used in the original derivation.
- All Ricci tensor computations were performed via direct Christoffel symbol and Riemann tensor summation (brute-force, no tensor-algebra package), cross-checked symbolically at each stage.

Appendix B: Reduction to RCPC via Areal Radius

This appendix shows explicitly that the free function C(r) appearing in the general solution (Section 4) is pure radial gauge freedom: an exact coordinate transformation maps the general-C(r) metric onto the standard RCPC form for any choice of C(r), not merely C(r)=1.

B.1 Definition of the areal radius

Define a new radial coordinate $\rho(r) = r\sqrt{C(r)}$, chosen so that the angular part of the metric reduces to $\rho^2 d\Omega^2$ with no leftover function in front — i.e., ρ is the areal (circumference) radius.

B.2 Differential relation

$d\rho/dr = \sqrt{C} + rC'/(2\sqrt{C}) = (2C + rC')/(2\sqrt{C})$, so $dr^2 = [4C/(2C+rC')^2]\, d\rho^2$.

B.3 Substitution into $B(r)dr^2$

Recall $B(r) = (rC'+2C)^2 / [4C(r^2|k|C+1)]$. Multiplying by dr^2 above, the factors $(rC'+2C)^2$ and 4C cancel identically, leaving $B(r)dr^2 = d\rho^2 / (r^2|k|C+1)$.

B.4 Rewriting in terms of ρ

Since $\rho^2 = r^2C$ by definition, $r^2|k|C + 1 = |k|\rho^2 + 1$, so $B(r)dr^2 = d\rho^2 / (1+|k|\rho^2)$. The angular part $C(r)r^2d\Omega^2 = \rho^2d\Omega^2$ by construction of ρ.

B.5 Result and conclusion

The full spatial line element becomes $dl^2 = d\rho^2/(1+|k|\rho^2) + \rho^2d\Omega^2$ — exactly the $B = 1/(1+kr^2)$ form from Section 5, now written in ρ for any choice of C(r), not just C(r)=1. Every trace of C(r) cancels identically; this is an exact algebraic identity, not an approximation or special case.

Combined with the Ricci-scalar invariant argument of Section 5, this establishes on two independent grounds that the free function $C(r)$ is pure radial gauge freedom, equivalent to the standard RCPC choice, and that the sign discrepancy is fully accounted for by the $k \to -k$ relabeling.

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